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Thermal processing susceptor

Abstract

In one embodiment, a susceptor for thermal processing is provided. The susceptor includes an outer rim surrounding and coupled to an inner dish, the outer rim having an inner edge and an outer edge. The susceptor further includes one or more structures for reducing a contacting surface area between a substrate and the susceptor when the substrate is supported by the susceptor. At least one of the one or more structures is coupled to the inner dish proximate the inner edge of the outer rim.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/183,146, filed Feb. 23, 2021, which is a continuation of U.S. patent application Ser. No. 16/109,945, filed Aug. 23, 2018, which is a divisional of U.S. patent application Ser. No. 14/698,793, filed Apr. 28, 2015, which claims benefit of U.S. Provisional Patent Application Ser. No. 62/001,562, filed on May 21, 2014, which are each herein incorporated by reference.

FIELD

(1) Embodiments of the present disclosure generally relate to a susceptor for thermal processing of semiconductor substrates, and more particularly to a susceptor having features to improve thermal uniformity across a substrate during processing.

BACKGROUND

(2) Semiconductor substrates are processed for a wide variety of applications, including the fabrication of integrated devices and microdevices. One method of processing substrates includes depositing a material, such as a dielectric material or a conductive metal, on an upper surface of the substrate. Epitaxy is one deposition process that is used to grow a thin, ultra-pure layer, usually of silicon or germanium on a surface of a substrate in a processing chamber. Epitaxy processes are able to produce such quality layers by maintaining highly uniform process conditions, such as temperature, pressures, and flow rates, within the processing chambers. Maintaining highly uniform process condition in areas around the upper surface of the substrate is necessary for producing the high-quality layers.

(3) Susceptors are often used in epitaxy processes to support the substrate as well as heat the substrate to a highly uniform temperature. Susceptors often have platter or dish-shaped upper surfaces that are used to support a substrate from below around the edges of the substrate while leaving a small gap between the remaining lower surface of the substrate and the upper surface of the susceptor. Precise control over a heating source, such as a plurality of heating lamps disposed below the susceptor, allows a susceptor to be heated within very strict tolerances. The heated susceptor can then transfer heat to the substrate, primarily by radiation emitted by the susceptor.

(4) Despite the precise control of heating the susceptor in epitaxy, temperature non-uniformities persist across the upper surface of the substrate often reducing the quality of the layers deposited on the substrate. Undesirable temperature profiles have been observed near the edges of the substrate as well as over areas closer to the center of the substrate. Therefore, a need exists for an improved susceptor for supporting and heating substrates in semiconductor processing.

SUMMARY

(5) In one embodiment, a susceptor for thermal processing is provided. The susceptor includes an outer rim surrounding and coupled to an inner dish, the outer rim having an inner edge and an outer edge. The susceptor further includes one or more structures for reducing a contacting surface area between a substrate and the susceptor when the substrate is supported by the susceptor, wherein at least one of the one or more structures is coupled to the inner dish proximate the inner edge of the

outer rim.

(6) In another embodiment, a susceptor for a thermal processing chamber is provided. The susceptor includes an outer rim surrounding and coupled to an inner dish, the outer rim having an inner edge and an outer edge. The susceptor further includes one or more elevated structures relative to an upper surface of the inner dish, the one or more elevated structures to reduce a contacting surface area between the susceptor and a substrate to be supported by the susceptor, wherein at least one of the elevated structures is coupled to the inner dish at a location proximate the inner edge of the outer rim.

(7) In another embodiment, a susceptor for a thermal processing chamber is provided. The susceptor includes an outer rim surrounding and coupled to an inner dish, the outer rim having an inner edge and an outer edge. The susceptor further includes six wedges extending radially inward from the inner edge of the outer rim above the inner dish, wherein each wedge is separated from two other wedges by a gap. The susceptor further includes a quartz insulating separator disposed between each of the wedges. Each quartz insulating separator contacting two wedges and the inner edge of the outer rim. The susceptor further includes three bumps extending from an upper surface of the inner dish. Each bump is located closer than each wedge to a center of the inner dish, wherein no bisection of the inner dish comprises all three bumps.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the embodiments disclosed above can be understood in detail, a more particular description, briefly summarized above, may be had by reference to the following embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments and are therefore not to be considered limiting of its scope to exclude other equally effective embodiments.

(2) FIG. 1A is a top sectional view of a susceptor, according to one embodiment.

(3) FIG. 1B is a top sectional view of a susceptor, according to another embodiment.

(4) FIG. 1C is a top sectional view of a susceptor, according to another embodiment.

(5) FIG. 1D is a top sectional view of a susceptor, according to another embodiment.

(6) FIG. 1E is a partial cross sectional view of a susceptor, according to the embodiment shown in FIG. 1D.

(7) FIG. 2A is a top sectional view of a susceptor, according to another embodiment.

(8) FIG. 2B is a partial cross sectional view of a susceptor, according to the embodiment shown in FIG. 2A.

(9) FIG. 2C is a top sectional view of a susceptor, according to another embodiment.

(10) FIG. 3A is a top sectional view of a susceptor, according to another embodiment.

(11) FIG. 3B is a partial cross sectional view of a susceptor, according to the embodiment shown in FIG. 3A.

(12) FIG. 4 is a partial cross sectional view of a susceptor, according another embodiment.

(13) FIG. 5 is a partial cross sectional view of a susceptor, according another embodiment.

(14) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

(15) The embodiments disclosed generally relate to a susceptor for thermal processing of semiconductor substrates. The embodiments disclosed can improve thermal uniformity across the surface of a substrate during processing by reducing a contacting surface area between the

susceptor and the substrate. Reducing the contacting surface area between the susceptor and the substrate reduces the amount of heat that is transferred from the susceptor to the substrate by conduction during processing. Embodiments of some structures that can reduce the contacting surface area between the substrate and the susceptor are described below.

(16) FIG. 1A is a top sectional view of a susceptor **120**, according to one embodiment. The susceptor **120** includes an outer rim **110** surrounding and coupled to an inner dish **102**. The inner dish **102** could be concave with the center of the inner dish being slightly lower than the edges of the inner dish **102**. The outer rim **110** includes an inner edge **112** and an outer edge **114**. Susceptors, such as susceptor **120**, are generally sized so that the substrate to be processed on the susceptor fits just inside the outer rim, such as the outer rim **110**. The susceptor **120** further includes lift pins **104** to aid in transferring substrates into and out of a thermal processing chamber (not shown) housing the susceptor **120**.

(17) The susceptor **120** further includes six wedges **122** for reducing a contacting surface area between a substrate (not shown) and the susceptor **120** when the substrate is supported by the susceptor **120**, wherein the wedges **122** contact the inner dish **102** proximate the inner edge **112** of the outer rim **110**. The wedges **122** may be formed as an integral part of the susceptor **120**, or may be attached to the susceptor, for example by welding. Each wedge **122** extends radially inward from the inner edge **112** of the outer rim **110** and each wedge is separated from two other wedges by a gap **124**. The gaps **124** correspond to areas of the underside of the substrate that will not contact the susceptor **120** allowing for more heat to be radiated from the susceptor **120** to the substrate during processing while reducing conductive heating at the substrate edge. Each wedge **122** is an elevated structure relative to an upper surface of the inner dish **102**. The wedges **122** can be symmetrically arranged around the center of the inner dish **102**. Each wedge **122** could contact the inner edge **112** of the outer rim **110** and each wedge **122** could have an upper surface higher than the upper surface of the inner dish **102**. The inner dish, outer rim as well as the wedges could be fabricated from silicon carbide, silicon carbide coated graphite, graphite coated with glassy carbon, or other materials with high thermal conductivity.

(18) Although six wedges **122** are shown in FIG. 1A, two or more wedges can be used in different embodiments. FIG. 1B shows a top sectional view of a susceptor **140** with eight wedges **142** separated by gaps **144**. FIG. 1C shows a top sectional view of a susceptor **160** with nine wedges **162** separated by gaps **164**. In some embodiments, additional wedges can improve thermal uniformity during processing by reducing the size of the individual surface areas on the susceptor (i.e., the top surface of each wedge) transferring heat to the substrate by conduction. Additional wedges can improve thermal uniformity at the edges of the substrate because there are more gaps where the substrate is not contacting another surface. This improved thermal uniformity helps prevent hotspots from forming along the edges.

(19) Susceptors **140** and **160** further include three rounded bumps **118** extending from the upper of the inner dish **102**. Each bump **118** is located closer than each wedge **142**, **162** to a center of the inner dish **102**. The bumps **118** could be fastened to inner dish **102** through a threaded connection or other common fastening means. The bumps **118** may be made of the same material as the susceptor, or a different material, and may be made from silicon carbide, or graphite coated with silicon carbide or glassy carbon. When substrates are supported around the edges, such as when a susceptor is used during processing, the center of the substrate can sag below the edges of the substrate. Susceptor dishes, such as the inner dish **102**, are often slightly concave to prevent portions of an underside of a sagging substrate from contacting the susceptor dish during processing. On the other hand, to create the highly uniform process conditions in epitaxy, the distance between the upper surface of the inner dish and the lower surface of the substrate is kept quite low, for example less than 0.25 mm. If the substrate contacts the dish, heat is transferred from the inner dish to the substrate by conduction and thermal uniformity may be reduced.

(20) The bumps **118** can be used to support a sagging substrate preventing contact between the

inner dish **102** and the substrate. The bumps **118** provide a contact surface area between the sagging substrate and the susceptor that is smaller than the surface area of the substrate that would contact the inner dish **102** absent the bumps **118**. The bumps **118** can be evenly distributed around the center of the inner dish **102** as shown in FIGS. **1B** and **1C**. In some embodiments, to ensure adequate support of a sagging substrate there could always be at least one bump **118** on a side of the inner dish **102**.

(21) FIG. **1D** is a top sectional view of a susceptor **180**, according to another embodiment. The susceptor **180** further includes six wedges **182** for reducing a contacting surface area between a substrate (not shown) and the susceptor **180** when the substrate is supported by the susceptor **180**, wherein the wedges **182** contact the inner dish **102** proximate the inner edge **112** of the outer rim **110**. Each wedge **182** extends radially inward from the inner edge **112** of the outer rim **110** and each wedge is separated from two other wedges by a gap **184**. The gap **184** is larger than the gap **124** shown in FIG. **1A** to further reduce the contacting surface area between the substrate and the susceptor. Susceptor **180** further includes an insulating separator **188** disposed between each of the wedges **182**.

(22) FIG. **1E** is a partial cross sectional view of the susceptor **180**, according to the embodiment shown in FIG. **1D**. The cross sectional view shows the top of the wedge **182** at the same height as the top of the insulating separator **188**. A substrate **50** is shown resting on the top of the wedge **182** and the insulating separator **188**. The insulating separator **188** may be disposed in a groove **189** formed around the susceptor surface near the inner edge **112** of the outer rim **110**, or at the specified radial location of the insulating separator **188**. The groove **189** maintains the insulating separator **188** in a specified location. The insulating separator **188** typically has a thickness that is greater than a depth of the groove **189** so an upper surface of the insulating separator **188** rises above the surrounding surface of the susceptor **220**, thus reducing contact between a substrate edge and the susceptor surface.

(23) The insulating separators **188** are typically made from a thermally insulating material, such as silicon oxide, quartz of any type (i.e. amorphous, crystalline, optical, bubble, etc.), glass, or the like. The insulating separators **188** provide thermal breaks, or areas of reduced thermal conductivity, around the inner dish during processing. This thermal break reduces thermal conduction into the edge of the substrate from the susceptor, which is typically made from a high thermal conductivity material. Reduced contact between the substrate edge and the highly conductive susceptor material reduces conductive heating of the substrate edge during processing. The insulating separators **188** may contact the inner edge **112** of the outer rim **110**, but could also be disposed at other locations on a susceptor. For example the insulating separators **188** may be spaced apart from the inner edge **112** of the outer rim **110**.

(24) FIG. **2A** is a top sectional view of a susceptor **220**, according to another embodiment. FIG. **2B** is a partial cross sectional view of the susceptor **220**. Referring to FIGS. **2A** and **2B**, the susceptor **220** is similar to the susceptor **120** including an outer rim **210** surrounding an inner dish **202**, the outer rim **210** having an inner edge **212** and an outer edge **214**. Three lift pins **204** can extend above the inner dish **202**.

(25) The susceptor **220** includes concentric annular ridges **222** surrounding the center of the inner dish **202**. Each annular ridge **222** has a different diameter. At least some of the annular ridges **222** can be located proximate the inner edge **212** of the outer rim **210**. In some embodiments, some of the annular ridges **222** may be located within about 1 mm of the inner edge **212**, for example within about 0.5 mm of the inner edge **212**. The susceptor **220** may further include six spokes **228** extending from the center of the inner dish **202** to the inner edge **212** of the outer rim **210**. More or fewer spokes **228** may be included in different embodiments. Each spoke **228** extends to a different angular location around the inner edge **212** of the outer rim **210**. In some embodiments, the upper surface of each spoke **228** is above the tops of the annular ridges **222**. In other embodiments, the upper surface of each spoke could be at substantially the same height as the tops of the annular

ridges **222**. In some embodiments, the annular ridges **222** continue under or through the spokes **228** making a complete ring around the center of the inner dish **202**. In other embodiments, the spokes **228** separate portions of the annular ridges **222**.

(26) The spokes **228** and annular ridges **222** can reduce a contacting surface area between a substrate **50** and the susceptor **220** when the substrate **50** is supported by the susceptor **220**. In some embodiments, the substrate **50** may only contact the spokes **228** during processing without contacting the annular ridges **222**. In other embodiments, the substrate **50** may contact both the spokes **228** and at least some of the annular ridges **222** during processing. In some embodiments, the one or both of the annular ridges **222** and the spokes **228** or their respective upper surfaces are elevated structures relative to the upper surface of the inner dish **202**. The ridges **222** may also improve thermal uniformity when processing a substrate by increasing radiative surface area of the upper surface of the susceptor **220**.

(27) The spokes **228** and annular ridges **222** may be made of the same material or a different material, which may be any of the same materials from which the susceptor **220** is made. The spokes **228** and annular ridges **222** may be made, in one embodiment, by sculpting the annular ridges **222** from an unpatterned susceptor dish surface. In another embodiment, recesses may be formed in an unpatterned susceptor dish surface to define the spokes, and then a pattern of ridged pieces attached to the susceptor surface within the recesses to form the annular ridges **222**, for example by welding.

(28) In some embodiments, the susceptor **220** can include an angled surface **216** connecting the inner dish **202** as well as the spokes **228** and annular ridges **222** to the inner edge **212** of the outer rim **210**. The angled surface **216** can be used as part of a supporting surface for the substrate **50**. Varying the slope or dimensions of the angled surface **216** can control the height of the substrate **50** relative to the spokes **228** and the annular ridges **222**.

(29) FIG. 2C is a top sectional view of a susceptor **240**, according to another embodiment. Susceptor **220** is similar to susceptor **240** except that susceptor **240** does not include any spokes **228**. Susceptor **240** includes annular ridges **242** that are similar to annular ridges **222**. When a substrate is placed on susceptor **240**, the underside of the substrate could contact at least some of the annular ridges **242** in some embodiments. In other embodiments, there could be a small gap between the underside of the substrate and the tops of the annular ridges **242** as the substrate is supported by a separate surface, such as angled surface **216** of FIG. 2B.

(30) FIG. 3A is a top sectional view of a susceptor **320**, according to another embodiment. FIG. 3B is a partial cross sectional view of the susceptor **320**. Referring to FIGS. 3A and 3B, the susceptor **320** is similar to the susceptor **120** including an outer rim **310** surrounding an inner dish **302**, the outer rim **310** having an inner edge **312** and an outer edge **314**. Three lift pins **304** can extend above the inner dish **302**.

(31) The susceptor **320** includes a series of bumps **322** extending from an upper surface of the inner dish **302**, so at least part of each bump is elevated above the inner dish **302**. At least some of the bumps **322** can be located proximate the inner edge **312** of the outer rim **310**. In some embodiments, some of the bumps **322** may be located within about 1 mm of the inner edge **312**, for example within about 0.5 mm of the inner edge **312**. Bumps **322** are arranged in a ringed pattern on the inner dish **302**, but other arrangements could be used, such as multiple rings, a triangular, square, or rectangular pattern, or a gridded pattern. In some embodiments, each quadrant of the inner dish **302** could include at least one bump **322**. The bumps **322** could be fastened to inner dish **302** through a threaded connection or other common fastening means.

(32) The bumps **322** can reduce a contacting surface area between a substrate **50** and the susceptor **320** when the substrate **50** is supported by the susceptor **320**. In some embodiments, the substrate **50** may only contact the bumps **322** during processing without contacting the inner dish **302** or any other surface. When the substrate **50** is supported using bumps **322**, hot spots around the edges of the substrates are greatly reduced. In other embodiments, additional bumps, such as bumps **118**

shown in FIGS. 1B and 1C could extend up from inner dish **302** at locations closer to the center of the inner dish **302** to prevent a sagging substrate from contacting the inner dish **302**.

(33) The bumps **322** are typically made from a low thermal conductivity material, such as silicon oxide, quartz of any type, glass, etc. The bumps provide a raised contact for the edge of a substrate disposed on the susceptor **320** to reduce conductive heating of the substrate edge. The bumps **322** may be inserted into recesses formed in the surface of the susceptor **320**. Features may be added to the bumps **322** and the recesses to allow the bumps **322** to be secured in the susceptor surface. Such features may include threads or other rotational engagement structures.

(34) FIG. 4 is a partial cross sectional view of a susceptor **420**, according another embodiment. The susceptor **420** is similar to the susceptor **120** including an outer rim **410** surrounding an inner dish **402**, the outer rim **410** having an inner edge **412** and an outer edge **414**. Three lift pins (not shown) could extend above the inner dish **402**.

(35) The susceptor **420** includes an annular ridge **422** extending from an upper surface of the inner dish **402**, so at least part of the annular ridge is elevated above the inner dish **402**. The annular ridge can surround the center of the inner dish **402** at a radial distance **424** from the center of the inner dish **402** that is less than the radius of a substrate **50** to be supported by the susceptor **420**. The annular ridge **422** may be made of a high thermal conductivity material, such as silicon carbide or graphite coated with silicon carbide or glassy carbon. A height **426** of the annular ridge **422** can be designed to control the gap between the substrate **50** and the inner dish **402**. In some embodiments, two or more annular ridges **422** could extend from the upper surface of the inner dish **402**. The additional annular ridges (not shown) could have different diameters as well as different heights from the other annular ridges. Annular ridge **422** is arranged in a ringed pattern on the inner dish **402**, but other arrangements could be used, such as multiple rings, a triangular, square, or rectangular pattern, or a gridded pattern. The annular ridge **422** can be located proximate the inner edge **412** of the outer rim **410**. In some embodiments, some of the annular ridges **422** may be located within about 1 mm of the inner edge **412**, for example within about 0.5 mm of the inner edge **412**.

(36) The annular ridge **422** can reduce a contacting surface area between a substrate **50** and the susceptor **420** when the substrate **50** is supported by the susceptor **420**. In some embodiments, the substrate **50** may only contact the annular ridge **422** during processing without contacting the inner dish **402** or any other surface. The radial location **424** as well as the height **426** of the annular ridge **422** could be modified to achieve different thermal profiles during processing. In other embodiments, bumps, such as bumps **118** shown in FIGS. 1B and 1C could extend from inner dish **402** at locations closer to the center of the inner dish **402** to prevent a sagging substrate from contacting the inner dish **402**.

(37) FIG. 5 is a partial cross sectional view of a susceptor **520**, according another embodiment. The susceptor **520** is similar to the susceptor **120** including an outer rim **510** surrounding an inner dish **502**, the outer rim **510** having an inner edge **512** and an outer edge **514**. Three lift pins (not shown) could extend above the inner dish **502**.

(38) The susceptor **520** includes an angled surface **522** extending radially inward from the inner edge **512** of the outer rim **510** to a depression **526**. At least part of angled surface **522** is an elevated structure relative to the upper surface of the inner dish **502**. The upper surface of the depression **526** is located below the upper surface of the inner dish **502**. The upper surface of the depression **526** couples the angled surface **522** to the upper surface of the inner dish **502**. The angled surface **522** could be angled between about three degrees and about twenty degrees from the upper surface of the inner dish **502**, such as between about four degrees and about twelve degrees, for example about seven degrees. The angle and location of the angled surface **522** can be used to control a radial location **524** corresponding to where the substrate **50** can contact the angled surface **522** during processing. The angle and location of the angled surface **522** can also be used to control the size of a gap **528** between the bottom of the substrate **50** and the upper surface of the depression

526. The size of the gap 528 could be between 0.1 mm and 1 mm, for example about 0.3 mm.

(39) The angled surface 522 can reduce a contacting surface area between a substrate 50 and the susceptor 520 when the substrate 50 is supported by the susceptor 520. In some embodiments, the substrate 50 may only contact the angled surface 522 during processing without contacting the inner dish 502 or any other surface. By using a relatively steep angle, such as between about three degrees and about twenty degrees from the upper surface of the inner dish 502, such as between about four degrees and about twelve degrees, for example about seven degrees, a smaller surface area of the substrate edge contacts the susceptor during processing, which reduces the amount of conductive heat that can be transferred from the susceptor 520 to the substrate 50. The angle and location of the angled surface 522 could be modified to achieve different thermal profiles during processing. In some embodiments, bumps, such as bumps 118, shown in FIGS. 1B and 1C could extend from inner dish 502 at locations closer to the center of the inner dish 502 to prevent a sagging substrate from contacting the inner dish 502.

(40) The susceptor embodiments described herein allow for more uniform temperature control of substrates during thermal processes, such as epitaxy. The temperature control is improved by reducing the surface area of the substrate contacting the susceptor, which reduces the amount of conductive heat transferred from the susceptor and the substrate. Conductive heat transfer between the susceptor and the substrate is more difficult to control than radiant heat transfer, the primary source of heat transfer between the susceptor and the substrate. Reducing the surface area of the substrate contacting the susceptor allows for a higher percentage of the heat transfer to be radiant heat resulting in improved temperature control and improved depositions on the substrate. The embodiments disclosed reduce the conductive heat transfer near the edge of the substrate by adding a structure, such as an annular ridge around the center of the inner dish proximate the outer rim, to reduce the contacting surface area between the susceptor and the substrate. The embodiments disclosed also prevent the possibility of substantial amounts of conductive heat transfer near the center of the substrate by including three bumps to support a substrate above the inner dish if the substrate sags.

(41) Although the foregoing embodiments have been described using circular geometries (e.g., inner dish, outer rim, annular ridge, etc.) to be used on semiconductor “wafers,” the embodiments disclosed can be adapted to conform to different geometries.

(42) While the foregoing is directed to typical embodiments, other and further embodiments may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method of processing a substrate comprising: positioning an outer portion of a substrate on: a plurality of wedges extending radially inward from an inner edge of an outer rim above an inner dish of a susceptor, wherein each wedge is separated from two other wedges by a gap and the outer rim is elevated relative to the plurality of wedges; and an insulating separator disposed between each of the wedges, each insulating separator contacting two wedges and the inner edge of the outer rim; and supporting an inner portion of the substrate with one or more bumps of a plurality of bumps, each bump located closer than each wedge to a center of the inner dish; performing a process on the substrate inside a process chamber.
2. The method of claim 1, wherein the plurality of bumps comprises at least three bumps.
3. The method of claim 1, wherein the plurality of bumps consists of three bumps.
4. The method of claim 3, wherein no bisection of the inner dish comprises all three bumps.
5. The method of claim 1, wherein the plurality of wedges comprises six wedges.
6. The method of claim 1, wherein the plurality of wedges comprises eight wedges.
7. The method of claim 1, wherein a top surface of each of the wedges is at a same height as a top

surface of each of the insulating separators.

8. The method of claim 1, wherein the insulating separators are quartz.

9. The method of claim 1, wherein a first inner sagging portion of the substrate is supported by a first bump of the plurality of bumps and a second inner sagging portion of the substrate is supported by a second bump of the plurality of bumps when performing the process on the substrate.

10. The method of claim 1, wherein each bump of the plurality of bumps is fastened to the inner dish.

11. The method of claim 1, wherein each bump of the plurality of bumps is formed of a same material as the inner dish of the susceptor.

12. A method of processing a substrate comprising: setting the substrate in a processing position on an inner dish of a susceptor disposed in a process chamber, the inner dish comprising an inner portion having a first upper surface and an outer depression having a second upper surface below the first upper surface, the setting comprising: positioning an outer portion of a substrate on a supporting structure of the susceptor, wherein the supporting structure extends to a first height above the first upper surface and the second upper surface of the inner dish of the susceptor; and supporting an inner portion of the substrate with one or more bumps extending from an upper surface of the inner dish of the susceptor, wherein a height of the one or more bumps is below the first height of the supporting structure; and performing a process on the substrate inside the process chamber when the outer portion of the substrate is positioned on the supporting structure and the inner portion of the substrate is supported by the one or more bumps, wherein the process performed on the substrate is a deposition of a layer on the substrate.

13. The method of claim 12, wherein the one or more bumps comprises at least three bumps.

14. The method of claim 12, wherein the one or more bumps consists of three bumps.

15. The method of claim 14, wherein no bisection of the inner dish comprises all three bumps.

16. A method of processing a substrate comprising: setting the substrate in a processing position on an inner dish of a susceptor disposed in a process chamber, the inner dish comprising an inner portion having a first upper surface and an outer depression having a second upper surface below the first upper surface, the setting comprising: positioning an outer portion of a substrate on a supporting structure of the susceptor, wherein the supporting structure extends to a first height above the first upper surface and the second upper surface of the inner dish of the susceptor; and supporting an inner portion of the substrate with one or more bumps extending from an upper surface of the inner dish of the susceptor, wherein a height of the one or more bumps is below the first height of the supporting structure; and performing a process on the substrate inside the process chamber when the outer portion of the substrate is positioned on the supporting structure and the inner portion of the substrate is supported by the one or more bumps, wherein a first inner sagging portion of the substrate is supported by a first bump of the one or more bumps and a second inner sagging portion of the substrate is supported by a second bump of the one or more bumps.

17. The method of claim 16, wherein the one or more bumps comprises at least three bumps.

18. The method of claim 16, wherein the one or more bumps consists of three bumps.

19. The method of claim 18, wherein no bisection of the inner dish comprises all three bumps.

20. The method of claim 16, wherein each bump of the one or more bumps is formed of a same material as the inner dish of the susceptor.
